



AALBORG UNIVERSITY
DENMARK

Written examination in the course
Engineering Optimisation - Concepts, Methods and Applications

Wednesday June 11th 2014

kl. 9.00 - 13.00 (4 hours)

All usual helping aids are allowed, i.e. books, notes, calculator, computer etc. All communication equipment and computer communication protocols must be turned off.

You may answer the questions in English or Danish.

REMEMBER to write your study number and page number on all sheets handed in.

The set consists of 7 exercises. The total weighting for each of the exercises is stated in percentage. You need 50 % in order to pass the exam.

It should be clear from the solution, which methods are used, and there should be a sufficient number of intermediate calculations, so the line of thought is clear.

Exercise 1: (25 %)

A company produces two products: 1) a MP3 player and 2) a Watch-TV (a wristwatch TV).

The production process for each product is similar in that both require a certain number of hours of electronic work and a certain number of labor-hours in the assembly department. Each MP3 player takes 4 hours of electronic work and 2 hours in the assembly shop. Each Watch-TV requires 3 hours in electronics and 1 hour in assembly.

During the current production period, 240 hours of electronic time are available, and 100 hours of assembly department time are available.

Each MP3 player sold yields a profit of 7 Euro; each Watch-TV produced may be sold for a 5 Euro profit.

- Formulate the problem of determining the best possible combination of MP3 players and Watch-TVs to manufacture to reach the maximum profit.
- Solve the problem formulated in a) using graphical methods.
- Set up the Lagrangian function for the problem formulated in a) and find point(s) satisfying the KKT necessary conditions. You may make use of results from b) when checking solving the KKT necessary conditions.

Exercise 2: (7 %)

The following multi-objective optimisation problem is considered:

$$\begin{aligned} \text{minimise} \quad & f_1(\mathbf{x}) = (x_1 - 8)^2 + (x_2 - 2)^2 + 3 \\ & f_2(\mathbf{x}) = (x_1 - 1)^2 + (x_2 - 7)^2 + 10 \end{aligned}$$

Figure 1 shows the Pareto optimal points in the design space and figure 2 shows the Pareto optimal set in the criterion space.

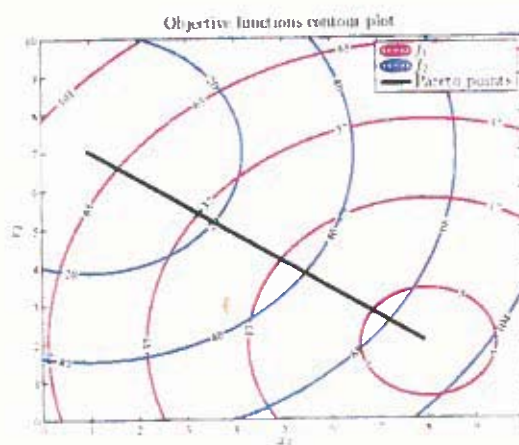


Figure 1: Contour curves and Pareto optimal points for the problem of exercise 3.

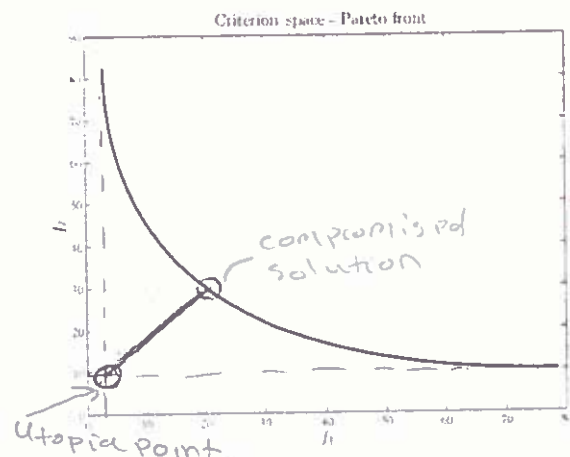


Figure 2: Pareto set in criterion space for the problem of exercise 3.

- Determine the objective function values of the utopia point.
- Assume that the multi-objective problem is solved as single objective problem using the weighting method (i.e. minimise $U(\mathbf{x}) = w_1 f_1(\mathbf{x}) + w_2 f_2(\mathbf{x})$, with $w_1 = w_2 = 1$). Determine the minimum objective function value $U(\mathbf{x}^*)$, and the optimum set of design variables $\mathbf{x}^*$.

Exercise 3: (10 %)

An optimisation problem is formulated as:

$$\text{minimise} \quad f(\mathbf{x}) = 4x_1^2 + 2x_2^2 - 2x_1x_2 + 2x_1 - x_2$$

Subject to:

$$\begin{aligned} g_1(\mathbf{x}) &= x_1 - 12 \leq 0 \\ g_2(\mathbf{x}) &= x_1^2 - x_1 - 2x_2 - 8 \leq 0 \\ g_3(\mathbf{x}) &= x_1 + x_2 - 20 \leq 0 \end{aligned}$$

The initial starting point for an optimisation is $\mathbf{x}^{(0)} = (-1, -2)$, for which the search direction is determined to $\mathbf{d}^{(0)} = [2 \ 7]^T$ and the vector of Lagrange multipliers for the constraints is $\mathbf{u} = [0 \ 0 \ 0]^T$. Let $R_0 = 1$ and $\gamma = 0.5$. Choose the trial step according to the sequence $t_0 = 1, t_1 = \frac{1}{2}, t_2 = \frac{1}{4}, t_3 = \frac{1}{8} \dots$

Calculate the step size using the inexact line search procedure (approximate step size procedure) and determine the new design variables, $\mathbf{x}^{(1)}$, for the next iteration.

Exercise 4: (8 %)

Solve the following problem by setting up the solution tree and using the Branch & Bound Method with Local Minimization:

$$\text{minimise} \quad f(\mathbf{x}) = -3x_1 - 8x_2$$

Subject to:

$$\begin{aligned} g_1(\mathbf{x}) &= -0.5x_1 + x_2 - 20 \leq 0 \\ g_2(\mathbf{x}) &= 2x_1 + x_2 - 20 \leq 0 \\ g_3(\mathbf{x}) &= 4x_1 - x_2 - 20 \leq 0 \end{aligned}$$

Both x_1 and x_2 should be integer values

As a help for solving the problem the objective function contours and constraints are plotted in figure 3.

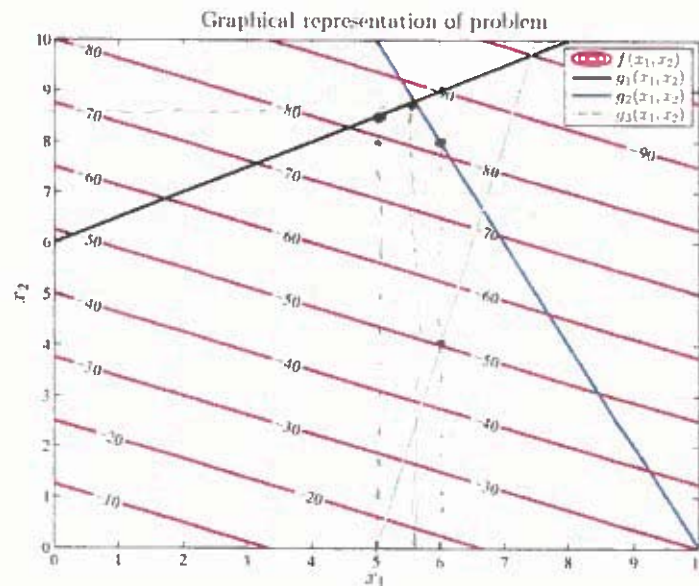


Figure 3: Graphical representation of problem in exercise 4.

Exercise 5: (20 %)

The cost function $f(\mathbf{x})$ is defined as the sum of squares of the residual function $r(\mathbf{x})$

$$f(\mathbf{x}) = \frac{1}{2} \sum_{i=1}^m r_i^2 \quad (1)$$

where

$$r_i(\mathbf{x}) = 2 + 2i - e^{(ix_1)} - e^{(ix_2)} \quad i = 1 \dots 10 \quad (2)$$

Note that the problem only holds two variables $\mathbf{x} = [x_1 \ x_2]$ ($n = 2$) and 10 points in the residual vector ($m = 10$).

The initial cost function $f(\mathbf{x}_0) \approx 2085.65$ for $\mathbf{x}_0 = [0.3, 0.4]$. The minimum value $f(\mathbf{x}^*) \approx 62.1811$ at $\mathbf{x}^* \approx [0.2578, 0.2578]$, see figure 4.

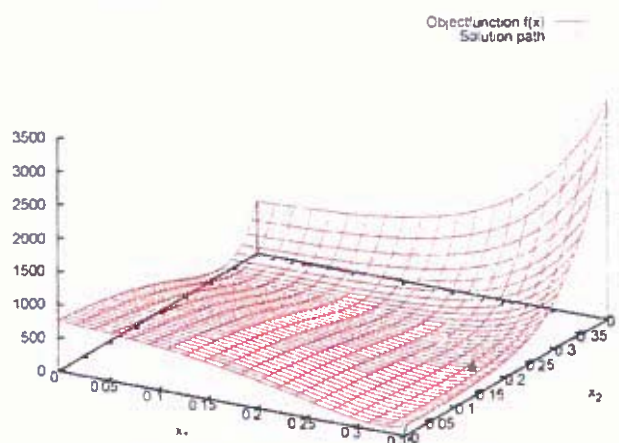


Figure 4: Surface plot of the cost function and the trajectory produced by an iterative solution scheme.

The following questions must be answered:

1. Derive an analytical expression for Jacobian matrix associated with equation 2.
2. Numerical solutions are expected for the following two sub-questions, where the initial parameters $\mathbf{x}_0 = [0.3, 0.4]$ are inserted into the expressions derived above.
 - (a) Using the initial parameters $\mathbf{x}_0 = [0.3, 0.4]$; calculate the gradient and approximate the Hessian using only first order information from the Jacobian matrix derived above in question 1.
 - (b) Complete the first 3 iterations using the Levenberg-Marquardt algorithm (where the step was defined as $\mathbf{s}_k = (\mathbf{H} + \lambda \mathbf{I})^{-1} \nabla f(\mathbf{x}_k)$). Start the iterations using a initial damping parameter $\lambda = 300$ and report λ , object-function $f(\mathbf{x}_k)$ and $\mathbf{x}_k$ for the first 3 iterations.

iter.	$\mathbf{x}$	$f(\mathbf{x})$	λ
0	$[0.3, 0.4]$	2085.7	300
1	$[0.3227, 0.2699]$	1466.3	600
2	$[0.2822, 0.2448]$	535.3	1200
3	$[0.3257, 0.2463]$	183.3	600

Exercise 6: (15 %)

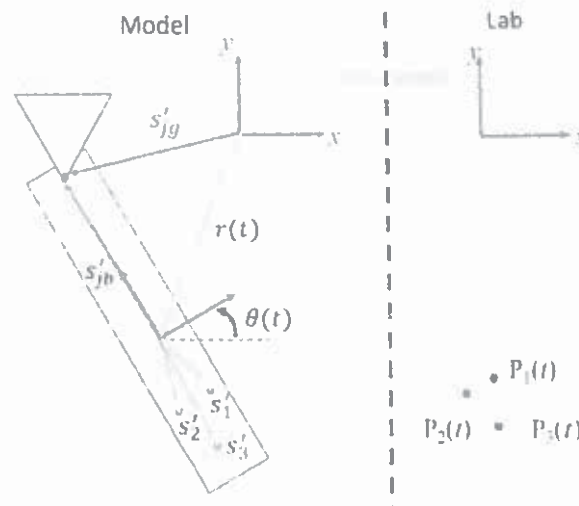


Figure 5: A rigid body connected to ground through a revolute joint.

Figure 5 shows a rigid body that is connected to ground through a revolute joint in a two-dimensional space. Three markers (the green dots in the figure) are attached to the body with local coordinates, s'_1 , s'_2 and s'_3 given in the indicated, body-fixed reference frame (indicated in red in the figure). The location of the revolute joint in body-fixed coordinates and in the global reference frame are given by s'_{jb} and s'_{jg} , respectively. The global reference frame is indicated in blue.

The position and orientation of the body is described in terms of a full cartesian formulation, i.e. position and orientation of the body relative to the global reference frames. The global position and rotation angle of the body are denoted, $r(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ and $\theta(t)$ as indicated on the figure.

A motion capture experiment was performed, where the position of the markers were measured in the global reference frame for a duration of time and the measured coordinates denoted $P_1(t)$, $P_2(t)$ and $P_3(t)$.

From the measured trajectories of the markers, it is desired to set up a nonlinear least-squares optimisation problem that can be used to compute the position and orientation of the rigid body. That is, the least-square difference between the measured markers and the points on the body is minimised while ensuring that the revolute joint constraint is fulfilled for each time step in the analysis.

The following questions must be answered:

1. Is the system of kinematic equations under-determinate, determinate or over-determinate when the motion of the rigid body connected to ground with a revolute joint is driven by the measured markers? Justify the answer by counting the number of degrees-of-freedom of the mechanical system compared to the number of kinematic driver equations.
2. Formulate a constrained nonlinear least-squares optimisation problem that can be used to compute both the position, $r(t)$ and rotation, $\theta(t)$, simultaneously.
3. Derive the first-order optimality conditions for the optimisation problem derived in question 2.

Exercise 7: (15 %)

In structural optimization we are often faced with min-max problems like minimizing the maximum stress in a structure.

- a) Explain the main mathematical difficulties of solving the problem of minimizing the maximum stress in a structure.*
- b) State the mathematical formulation for three fundamentally different ways of solving the optimization problem. Describe advantages and disadvantages of the three chosen optimization problem formulations.*

Exercise 1

- a) Formulate the problem of determining the best possible combination of MP3 players and watch-TVs to manufacture to maximize profit.

Info: 1 MP3 Player and Watch-TV

- MP3 player takes 4 hours of electronic work and 2 hours in the assembly shop
- Watch takes 3 hours of electronic work and 1 hour in assembly
- 240 hours of electronic work, and 100 hours of assembly are available.
- Profit for MP3 player is 7 euro
- Profit for watch is 5 euro.

Define design variables

n_{MP3} : number of MP3 players

n_W : number of watches

Standard optimization Design model

minimize : $f(n_{MP3}, n_W) = \frac{1}{n_{MP3} \cdot 7 + n_W \cdot 5}$

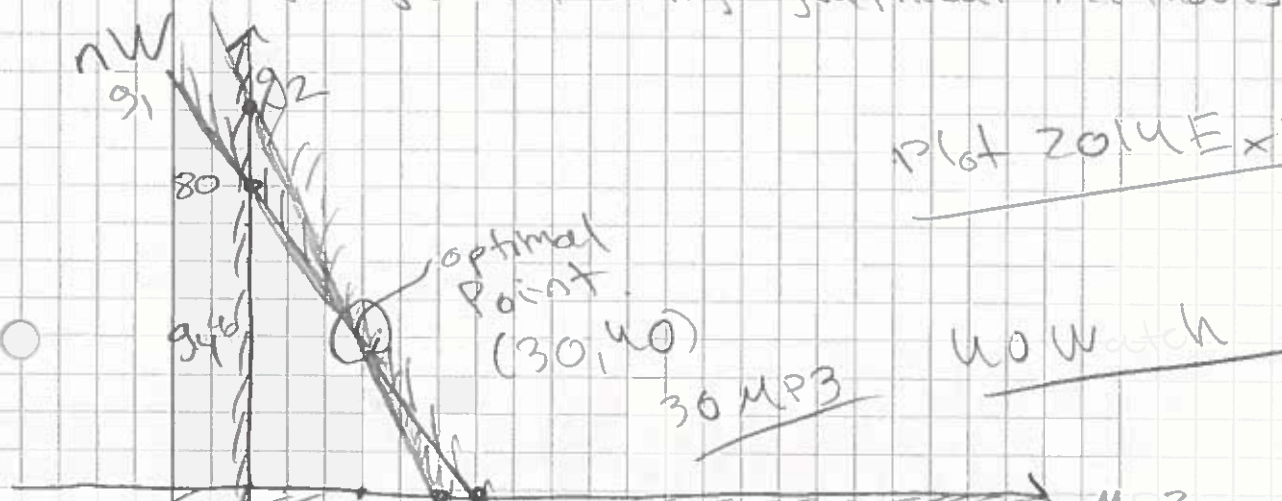
s.t. : $g_1: n_{MP3} \cdot 4 + n_W \cdot 3 \leq 240$
 $\Rightarrow n_{MP3} \cdot 4 + n_W \cdot 3 - 240 \leq 0$

$g_2: n_{MP3} \cdot 2 + n_W \cdot 1 \leq 100$
 $\Rightarrow n_{MP3} \cdot 2 + n_W \cdot 1 - 100 \leq 0$

$g_3: n_{MP3} \geq 0 \Rightarrow -n_{MP3} \leq 0$

$g_4: n_W \geq 0 \Rightarrow -n_W \leq 0$

- b) Solve the problem using graphical methods



c) Set up the Lagrangian function for the problem in a) and find point(s) satisfying the KKT necessary conditions. You may make use of results from b) when checking/solving the KKT necessary conditions.

Use Slide 18 Lecture 2 $MP_3 = x_1$ $W = x_2$

$$L(x, v, u, s) = f(x) + \sum_{j=1}^p v_j h_j(x) + \sum_{i=1}^m u_i (g_i(x) + s_i^2)$$

$$= -7 \cdot x_1 - 5 \cdot x_2 + u_1 (4 \cdot x_1 + 3 \cdot x_2 - 240 + s_1^2) + u_2 (2 \cdot x_1 + x_2 - 100 + s_2^2)$$

$$\frac{\partial L}{\partial x_1} = 4 \cdot u_1 + 2 \cdot u_2 - 7 = 0$$

$$\frac{\partial L}{\partial x_2} = 3 \cdot u_1 + u_2 - 5 = 0$$

$$\frac{\partial L}{\partial u_1} = s_1^2 + 4 \cdot x_1 + 3 \cdot (x_2 - 80) = 0$$

$$\frac{\partial L}{\partial u_2} = s_2^2 + 2 \cdot x_1 + x_2 - 100 = 0$$

$$\frac{\partial L}{\partial s_1} = 2 \cdot s_1 \cdot u_1 = 0$$

$$\frac{\partial L}{\partial s_2} = 2 \cdot s_2 \cdot u_2 = 0$$

Solve

$$s_1 = 0$$

$$s_2 = 0$$

$$u_1 = \frac{3}{2}$$

$$u_2 = \frac{1}{2}$$

$$x_1 = 30$$

$$x_2 = 40$$

Exercise 2.1

The following multi objective optimisation is considered.

$$\text{Minimise } f_1(x) = (x_1 - 8)^2 + (x_2 - 2)^2 + 3$$

$$f_2(x) = (x_1 - 1)^2 + (x_2 - 7)^2 + 10$$

Figure 1 shows the pareto optimal points in the design space and figure 2 shows pareto optimal set in the criterion space.

a) Determine the objective function values of the utopia point

(3, 10) seen from
graph
↑ ↑
 f_1 f_2

The coordinates of the pareto points used in the objective function

Point	f_1	f_2
$P_1(1, 7)$	77	10
$P_2(8, 2)$	3	84

$$\Rightarrow \text{utopia } (\min(f_1), \min(f_2)) = (3, 10)$$

b) Assume that the multi objective problem is solved as single objective problem using the weighting method (i.e. minimize $U(x) = w_1 f_1(x) + w_2 f_2(x)$ with $w_1 = w_2 = 1$). Determine the minimum objective function value $U(x^*)$ and the optimum set of design variables x^*

$$U(x) = 1 \cdot f_1(x) + 1 \cdot f_2(x)$$

$$= (x_1 - 8)^2 + (x_2 - 2)^2 + 3 + (x_1 - 1)^2 + (x_2 - 7)^2 + 10$$

The solution is the point that gives the shortest distance from the utopia point to the pareto front

That also equals to the center of the line in figure 1

1

$$\Rightarrow x_1^* = 1 + \frac{8-1}{2} = 4.5 \quad \text{and} \quad x_2^* = 2 + \frac{7-2}{2} = 4.5$$

$$x^* = \begin{bmatrix} 4.5 \\ 4.5 \end{bmatrix}$$

$$U(x^*) = f_1(x) + f_2(x) = 21.5 + 28.5 = 50$$

An optimization problem is formulated as

$$\text{Minimize: } f(x) = 4x_1^2 + 2x_2^2 - 2x_1x_2 + 2x_1 - x_2$$

s.t.

$$g_1(x) = x_1 - 12 \leq 0$$

$$g_2(x) = x_1^2 - x_1 - 2x_2 - 8 \leq 0$$

$$g_3(x) = x_1 + x_2 - 20 \leq 0$$

The initial starting point for an optimization is $x^{(0)} = (-1, -2)$, for which the search direction is determined to $d^{(0)} = [2 \ 7]^T$ and the vector of Lagrange multipliers for the constraints is $u = [0 \ 0 \ 0]^T$. Let $P_0 = 1$ and $\gamma = 0.5$. Choose the trial step according to the frequency $t_0 = 1, t_1 = \frac{1}{2}, t_3 = \frac{1}{8} \dots$

Calculate the step size using the exact line search procedure and determine the new design variables $x^{(1)}$ for the next iteration.

$$\Phi_0 = f_0 + P_0 \cdot V_0 \quad \text{Initial descent function}$$

Gather info for Φ_0

$$V_0 = \max \{0, |h_i|, g_i\}$$

$$f_0 = f(-1, -2) = 8$$

$$P_0 = 1$$

$$V_0 = \max \{0, -13, -2, -23\}$$

$$V_0 = 0$$

$$g_1(-1, -2) = -13 < 0 \quad \text{all inactive}$$

$$g_2(-1, -2) = -2 < 0$$

$$g_3(-1, -2) = -23 < 0$$

$$r_0 = \sum V_i + \sum u_i = 0 + 0 + 0 = 0$$

$$R = \max \{r_0, P_0\} = \max \{0, 1\} = 1$$

Penalty parameter

Plug in

$$\Phi_0 = 8 + 1 \cdot 0 = 8$$

Trial 1 $t_0 = 1$ and $j = 0$

$$\begin{bmatrix} x_1^{(1,0)} \\ x_2^{(1,0)} \end{bmatrix} = \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix} + t_0 \begin{bmatrix} d_1^{(0)} \\ d_2^{(0)} \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

Cost and constraint violations at the trial point

$$f_{1,0} = f(1, 5) = 41; \quad g_1(1, 5) = -11; \quad g_2(1, 5) = -18; \quad g_3(1, 5) = -14$$

All inactive $\Rightarrow V_0 = 0$

$$\Phi_{2,0} = f_{1,0} + R V_{1,0} = 41 + 1 \cdot 0 > \Phi_0$$

$\Rightarrow$ descent condition violated and we continue to next trial.

$$\begin{bmatrix} x_1^{(1,1)} \\ x_2^{(1,1)} \end{bmatrix} = \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \end{bmatrix} + t_1 \begin{bmatrix} d_1^{(0)} \\ d_2^{(0)} \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} + 0,5 \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ 1,5 \end{bmatrix}$$

Cost and constraint functions at the trial point.

$$f_{1,1} = f(0, 1,5) = 3, \quad g_1 = -12, \quad g_2 = -11, \quad g_3 = -18,5$$

all inactive $\Rightarrow V_{1,1} = 0$

Descent function

$$\Phi_{1,1} = f_{1,1} + R V_{1,1} = 3 + 0 = 3 < \Phi_0 = 8 \Rightarrow \text{not violated}$$

we check for further requirements

$$\beta_1 = 8 \|d^{(0)}\|^2 = 0,5 \cdot (2^2 + 7^2) = 26,5$$

$$\Phi_{1,1} + t_1 \cdot \beta_1 = 3 + 0,5 \cdot 26,5 = 16,25 > \Phi_0 = 8 \Rightarrow \text{Violated}$$

Trial 3 $t = 0,25$ $j = 2$

$$\begin{bmatrix} x_1^{(1,2)} \\ x_2^{(1,2)} \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} + 0,25 \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} -0,5 \\ -0,25 \end{bmatrix}$$

Cost and constraint functions at the trial point

$$f_{1,2} = 0,125, \quad g_1 = -12,5, \quad g_2 = -6,75, \quad g_3 = -20,75$$

all inactive $\Rightarrow V_{1,2} = 0$

$$\Phi_{1,2} = f_{1,2} + R \cdot V_{1,2} = 0,125 + 1 \cdot 0 = 0,125 < \Phi_0 = 8 \Rightarrow \text{not violated}$$

check further

$$\Phi_{1,2} + t_2 \cdot \beta_1 = 0,125 + 0,25 \cdot 26,5 = 6,75 < \Phi_0 = 8 \Rightarrow \checkmark$$

The descent condition is fulfilled and the new design variables are

$$x^{(1)} = \begin{bmatrix} -0,5 \\ -0,25 \end{bmatrix}$$

and step size $t_2 = 0,25$

Exercise 1

Solve the following problem by setting up the solution tree and using the Branch and Bound Method with local minimization

$$\text{minimize } f(x) = -3x_1 - 8x_2$$

s.t.

$$g_1(x) = -0.5x_1 + x_2 - 20 \leq 0$$

$$g_2(x) = 2x_1 + x_2 - 20 \leq 0$$

$$g_3(x) = 4x_1 - x_2 - 20 \leq 0$$

We have figure on exam

Continuous guess of solution

initial guess

$$x_c^* = \begin{bmatrix} 5.5 \\ 8.5 \end{bmatrix} \Rightarrow f(x_c^*) = -84.5$$

$$x_1 \leq 5$$

$\pm x_1$ or vel x_2 at
graph method not
bad.

$$x_1 \geq 6$$

$$x_c^* = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$$

$$\Rightarrow f(x_c^*) = -82$$

$\pm$

Optimum

$$x_c^* = \begin{bmatrix} 5 \\ 8.3 \end{bmatrix}$$

$$\Rightarrow f(x_c^*) = -81.4$$

$$x_2 \leq 8$$

$$x_c^* = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$$\Rightarrow f(x_c^*) = -79$$

$$x_2 \geq 9$$

not feasible

$\pm$

The function $f(x)$ is defined as the sum of squares of the residual function $r(x)$

$$f(x) = \frac{1}{2} \sum_{i=1}^m r_i^2 \quad (1)$$

where

$$r_i(x) = 2 + 2 \cdot e^{(i \cdot x_1)} - e^{(i \cdot x_2)} \quad i = 1, \dots, 10 \quad (2)$$

Note that the problem only holds two variables $x = [x_1 \ x_2]$ ($n=2$) and 10 points in the residual vector ($m=10$).

The initial cost function $f(x_0) \approx 2085.65$ for $x_0 = [0.3 \ 0.4]$
The minimum value $f(x^*) \approx 62.1811$ at $x^* \approx [0.2578 \ 0.2578]$

See figure in exam.

1. Derive analytically expression for Jacobian matrix associated with equation (2)

$$J = \begin{bmatrix} \frac{\partial r_1(x)}{\partial x_1} & \frac{\partial r_1(x)}{\partial x_2} \\ \frac{\partial r_2(x)}{\partial x_1} & \frac{\partial r_2(x)}{\partial x_2} \\ \vdots & \vdots \\ \frac{\partial r_{10}(x)}{\partial x_1} & \frac{\partial r_{10}(x)}{\partial x_2} \end{bmatrix} = \begin{bmatrix} -e^{x_1} & -e^{x_2} \\ -2 \cdot e^{2 \cdot x_1} & -2 \cdot e^{2 \cdot x_2} \\ -3 \cdot e^{3 \cdot x_1} & -3 \cdot e^{3 \cdot x_2} \\ \vdots & \vdots \\ -10 \cdot e^{10 \cdot x_1} & -10 \cdot e^{10 \cdot x_2} \end{bmatrix}$$

2. Numerical solutions are expected for the following two sub-questions, where the initial parameters $x_0 = [0.3 \ 0.4]$ are injected into the expression derived above.

a) Using the initial parameters: Calc gradient and approx Hessian using only 1st order info

• Gradient: $\nabla f(x) = J^T \cdot r$

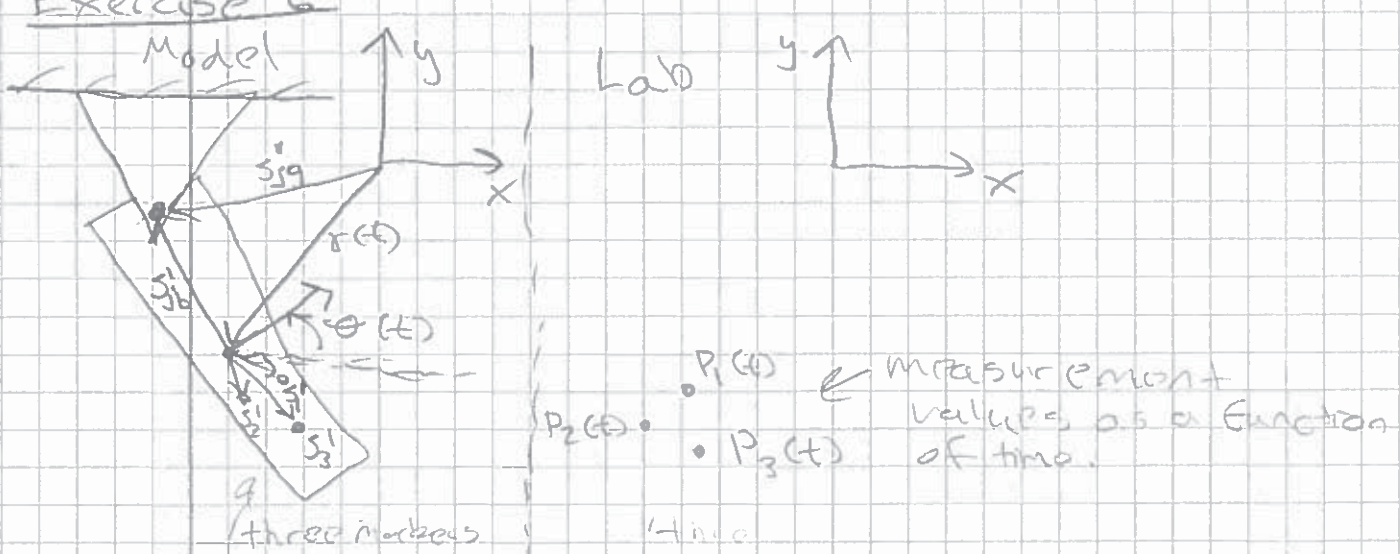
$$= \begin{bmatrix} -e^{0.3} & -2 \cdot e^{2 \cdot 0.3} & \dots & -10 \cdot e^{10 \cdot 0.3} \\ -e^{0.4} & -2 \cdot e^{2 \cdot 0.4} & \dots & -10 \cdot e^{10 \cdot 0.4} \end{bmatrix} \cdot \begin{bmatrix} 2 + 2 \cdot e^{0.3} - e^{0.4} \\ 2 + 2 \cdot e^{2 \cdot 0.3} - e^{2 \cdot 0.4} \\ \vdots \\ 2 + 2 \cdot e^{10 \cdot 0.3} - e^{10 \cdot 0.4} \end{bmatrix}$$

$$= \begin{bmatrix} 1.6898 \cdot 10^4 \\ 4.3701 \cdot 10^4 \end{bmatrix}$$

• Hessian: $J^T \cdot J = \begin{bmatrix} -e^{0.3} & -2 \cdot e^{2 \cdot 0.3} & \dots & -10 \cdot e^{10 \cdot 0.3} \\ -e^{0.4} & -2 \cdot e^{2 \cdot 0.4} & \dots & -10 \cdot e^{10 \cdot 0.4} \end{bmatrix} \cdot \begin{bmatrix} -e^{0.3} & -2 \cdot e^{2 \cdot 0.3} & \dots & -10 \cdot e^{10 \cdot 0.3} \\ -e^{0.4} & -2 \cdot e^{2 \cdot 0.4} & \dots & -10 \cdot e^{10 \cdot 0.4} \\ \vdots & \vdots & \vdots & \vdots \\ -10 \cdot e^{10 \cdot 0.3} & -10 \cdot e^{10 \cdot 0.4} \end{bmatrix} \cdot 10$

$$= \begin{bmatrix} 0.7139 & 1.8124 \\ 1.8124 & 4.6461 \end{bmatrix}$$

b) First 3 iterations in L-M see exam



Rigid body connected to ground through a revolute joint

The global position and rotation angle of the body are denoted, $r(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ and $\theta(t)$

- The least-squares difference between the measured markers and the points on the body is minimized while ensuring that the revolute joint constraint is fulfilled for each time step in the analysis.

1/ Is the system of kinematic equations under-determined or over-det. when the motion is driven by the measured markers?

$$\text{DOF} = 3 \cdot \text{Links} - 2 \cdot \text{joints} - 3 \cdot \text{grounded}$$

$$M = 3 \cdot 2 - 2 \cdot 1 - 3 \cdot 1 = 1 \text{ DOF}$$

↑
bodies + ground

3 drivers > 1 DOF $\Rightarrow$ over determined

2/ Formulate a constrained nonlinear least-squares optimization problem that can be used to compute both the position $r(t)$ and rotation, $\theta(t)$ simultaneously.

Position analysis: revolute joint: $\Phi_0 = s_{1g} - (r(t) + A(\theta(t)) \cdot s_{1b}) = 0$

rotation matrix local system $\Rightarrow$ rotate to global

$$r = \begin{bmatrix} r(t) + A(\theta(t)) \cdot s_1 - P_1(t) \\ r(t) + A(\theta(t)) \cdot s_2 - P_2(t) \\ r(t) + A(\theta(t)) \cdot s_3 - P_3(t) \end{bmatrix}$$

Residual

and least squares:

$$f = \frac{1}{2} \sum \|r\|^2$$

minimize $f = \frac{1}{2} \sum \|r\|^2$

s.t. $\Phi = 0$

$$A = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$$

5) Derive the first order optimality conditions for the optimization problem derived in question 2

$$L = f(x) + \underbrace{\sum V h(x)}_{\text{equality on constraints}} \quad (L = f + h^T \lambda)$$

$$= \frac{1}{2} ((r(t) + A(\theta(t)) \cdot s_1 - p_1)^2 + (r(t) + A(\theta(t)) \cdot s_2 - p_2)^2 + (r(t) + A(\theta(t)) \cdot s_3 - p_3)^2 + V_1 \cdot (s_{jg} - (r(t) + A(\theta(t)) \cdot s_{j0})))$$

and $\frac{\partial L}{\partial x_k} = 0$; $\frac{\partial L}{\partial v_j} = 0$ (we know ∇h - Jacobian)
 $L \lambda = h = 0$ (row in it)

$$L_x = f_x^T + h_x^T \lambda = 0$$

$$KKT: G_q^T + \Phi_q^T \lambda = 0$$

$$\begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

$$\Phi = 0$$

$$\text{Jacobian} \begin{bmatrix} x & y & \theta \\ x & y & \theta \end{bmatrix}$$

See pic from Martin.

Loc 11 slide 32

Exercise 7

min-max optimization problem

See 2015 exam.

